

## Behavioral/cognitive description of autism : some “modeling theories”

- deficit in executive functions

### Focus on details / integration / complexity

- weak central coherence : cognitive style focused on details -> no global vision or context
  - implicit language : necessitates integration of several info sources + context
  - social : no comprehension of social situations in their globality (too focus on one word etc)
  - limited interest / attraction to regularities
  - initial version : inability to integrate info, later version : local processing bias
- baron-cohen 2006: hyper-systemizing theory of autism proposes that the systemizing mechanism is set too high in people with autism
  - difficulty to cope with high-variance systems
  - // wcc but for positive reasons : serves a purpose (understanding a system)
- monotropism murray2005 : low range of interests // wcc
  - competition for attention shapes cognitive processes
  - lack of generalization
    - \* top-down processing strict/limited/resistent to correction
    - \* hyperawareness within the attentinal tunnel, hypoawareness outside it
- dwyer2024, supports monotropism account of ASD: strong attention for one subject, but difficulty to divide att
  - atypical regulation of attention rather than deficit, which might manifest differently in different contexts
  - ASD : both increased endogenous hyper-focus and increased susceptibility to exogenous attention capture (inattention)
  - more hyperfocus correlated with less quality of life (QOL)
    - \* but positive effect on QOL mediated by positive engagement in the hyper-focus activity
- ben-yosef2016
  - compares 3 accounts : weak central coherence (local/global), complexity and social cognition deficit
  - no deficit in processing human stimuli, nor in global processing (conceptual or perceptual)
    - \* despite previous reported findings of deficits ???
  - various studies showed intact context processing in direct associative context (image and sound of the dog)
    - \* but impaired integrative context (none of the cues directly leads to the target, necessitates integration)
  - CCL : diff when synchronization between different regions is required, but NOT ONLY, also cognitive overload in general
    - \* supports complexity account of ASD
- motttron 2000
  - cognitive deficits in unrelated areas (e.g., visuospatial, music, language, memory) in ASD
    - \* hyp : multi-modal deficit ? (several sensory modalities, or several cognitive operations in the same modality)
    - \* or WCC : deficit in integration into a higher level of organization ? would result in focus on details
  - hierarchisation deficit hypothesis (HDH) : local and global function normally, but global has no advantage over local
  - same/different judgement with local (note change)/global (transposed) change in melodies
    - \* no impaired global processing, but local advantage!
    - \* enhanced perception for elementary properties of sounds rather than deficit in integration ??
  - Paradoxical Functional Facilitation : how neurological deficit may induce enhanced perf -> P compensates for non-P processes ?

### Atypical sensorial processing

- overreactivity as a primary symptom in autism (Liss, 2006)
  - hyp : ind with ASD suffer from overarousal and this primary diff may lead to overselective attention and repetitive behavior.
  - cluster analysis : subgroup with both overreactivity and overselective attention
    - \* overfocusing cluster highest functioning but most socially impaired -> OF detrimental to social behavior, not to intellect ?
    - \* lowest-functioning cluster more underreactive -> linked to mental retardation ?
  - hyp: overselective attention/repetitive behavior compensation mechanisms -> predictable/repetitive events moderate arousal level
  - overarousal hyp reformulated into “noisy firing of neurons”
    - \* hussmann (2001) suggests lack of inhibition (gaba-ergic neurons)
    - \* Casanova 2002 : microcellular columns in the brain (post-mortem) smaller and more numerous -> overconnectivity, no selectivity
    - \* Kinsbourne attributed autistic symptomatology to exaggerated ascending activation of the cortex from the brain-stem.

- gomes 2008 hypersensitivity : some authors had also correlated the sensorial abnormalities with attention, demonstrating that the fact of being very reactive is related to attention increase, and vice versa
- hilton 2007 link sensory/social profile
  - Dunn's (1997) model of sensory processing : low/high threshold and active/passive response
  - moderate to strong relationship sensory response / social scale
  - mostly for
    - \* sensation avoiding (low threshold, active) : afraid of being overwhelmed during social interaction ?
    - \* sensory sensitivity (low threshold, passive) : easily distracted/hyperactive, miss social cues ?
  - moderately to
    - \* sensation seeking (high threshold, active) : attention to SS or odd behavior discourage other children
    - \* low registration (high threshold, passive) : do not notice sensory info from the environment -> miss social cues ?

## Empathy

- milton 2012 : double empathy problem -> non-ASD lack empathy and understanding of ASD people
- crompton2020 : “téléphone arabe” : ASD and non-ASD are effective, mixed chains are less effective
  - Effective information transfer requires social communication skills. As autism is clinically defined by social communication deficits, it may be expected that information transfer between autistic people would be particularly deficient. However, the Double Empathy theory would suggest that communication difficulties arise from a mismatch in neurotype; and thus information transfer between autistic people may be more successful than information transfer between an autistic and a non-autistic person.
- alkhaldi2021 : neuro-typical dislike more autistic people, they find them awkward and with a lack of empathy
- dunn2023 : autistic people rate themselves as awkward but with appeal to socialize with them -> not universally “undesirable”

## General model for hypotheses on autism (inspired from Rubenstein2003)

1. Precoce genetic or environmental alteration (gene mutation or atypical environment)
2. Less GABAergic neurons in cortex/amygdale (less produced or migrated)
3. Less inhibition (GABA) compared to excitation (glutamate) = déséquilibre excitation/inhibition
4. Excessive Plasticity (too much LTP, freezed circuits) -> learning too fast and less flexible, overreactivity to stimuli
5. underdifferentiated representations, simple stimuli activate large populations of neurons
6. Local excess excitability -> less differentiation -> high distance lack of excitability -> lower connectivity
7. Atypical behavior like in autism (hypersensitivity, rigid memory, less flexibility, increased risk of epilepsy)

## Other

- Female Autism Phenotype (FAP) hull2020
  - higher social motivation, but difficulty to maintain long-term relationship
  - camouflaging : use of conscious or unconscious strategies, which may be explicitly learned or implicitly developed, to minimise the appearance of autistic characteristics during a social setting
    - \* more difficult to diagnose, often misdiagnosed as anxiety etc
    - \* strong associations between camouflaging and poor mental health outcome

## Findings cognitive tests/behavioral measures

- lau 2012 : high prevalence of ASD among AgCC, but ASD shared with many parents, AgCC not
  - AgCC intermediate scores between control and ASD -> one part of the story ??
- Wheelwright2010 : Autism Spectrum Quotient (AQ)
  - NAP : narrow autism phenotype (>3sd)
  - BAP : broad AP (>1sd)
  - MAP : medium AP (>2sd)
  - more males than females, more scientists, in particular mathematiciens, high prevalence on parents

## Findings about neuro correlates ASD

- piven1997: smaller CC, bigger cortical regions, especially for parietal
  - smaller only when control for brain size -> maybe only bigger cortical regions, esp. where no connection with the CC
- Just2006 : underconnectivity fronto-parietal, may come from anatomical constraints (smaller genu CC)

- Casanova 2002 : more short fibers than longer -> deficit in long-distance connectivity
- executive dysfunction could come from underconnectivity fronto-parietal
- but normal activation of each region -> diff tasks requiring integration of different regions
  - \* theory of mind, face processing, executive functions could be all result of integrating info
  - \* postural deficits ?? Minshew 2004
- CCL: It would be particularly interesting to examine functional connectivity in participants with complete or partial agenesis of the corpus callosum but who nevertheless show relatively unimpaired cognitive functioning to determine how callosal absence affects interhemispheric functional connectivity and presumably, the degree of coordination between hemispheres.
- mason2008:
  - 3 types of inferences: intentional (ToM-based), physical, and emotional
  - ASD : + RH activation during reading during 3 types of inference, as well as during context sentence
    - \* hyp : spillover : RH recruited because LH is overloaded
  - reduced functional connectivity :
    - \* within the ToM network (between left medial frontal gyrus and right temporo-parietal junction)
    - \* btwn ToM and LH language network (left inferior frontal gyrus, and left superior temporal gyrus)
    - \* FC between frontal/parietal correlated with size of the CC genu
- Alexander 2007 : only a subgroup of autists had decreased FA/increased meandiffusivity driven by increase radial diffusivity
  - linked to a default in myelinisation ? + correlation with sizeof CC and processing speed
  - Similar conclusions in Shukla2010
- Belmonte2004 :
  - ASD : high local connectivity may develop in tandem with low long-range connectivity (Just et al., 2004)
  - overconnected : everyone surpasses threshold in sensory areas -> no selectivity in sensory areas -> overload in next steps (no distinction target/non-target) -> integratory areas get no normal input -> low activation
- Belmonte2004 : cerebellum : Particularly implicated in deficits of long-range connectivity and coordination of cognitive functions
  - ASD : smaller cerebellum, fewer Purkinje cells that handle inhibition -> high physical conn, low computational connectivity
    - \* link with cognitive, motor, social deficits
- Rubenstein2003 : Modulated noise exposure model of hyper-sensibility
  - critical window to create sound map/calibration of equil exc/inhib -> generates end of critical period elsewhere in the cortex
    - \* modulated noise -> closes early with no differentiation because poor signal/noise ratio
    - \* continuous noise -> never closes, because no closure signal
    - \* GABAergic neurons : lateral inhibition (between cortical columns), intra-column activation (prevent excessive activation), regulation timing synaptic plasticity
- General model for hypotheses on autism (inspired from Rubenstein2003)
  - 1. Precoce genetic or environmental alteration (gene mutation or atypical environment)
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  - 3. Less inhibition (GABA) compared to excitation (glutamate) = déséquilibre excitation/inhibition
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- Depretto 2005 : no activation of Mirror Neuron System (MNS) during observation/imitation emotional expressions
  - tasked performed as well as controls/no difference in fixation duration of eyes/faces
  - relying on visual/motor attention
  - activity NMS correlated with social ability
- diamond 2019 : fixel-based analysis ASD : decreased FD in CC in the splenium/genu -> reduced axonal density ?
  - decreased GWM-FD (global white matter fiber density), même si plus petite diff que le CC -> generalized disorder ?
  - negative correlation social impairment / fiber density in the CC splenium
- paul 2012 hyp:
  - When a brain loses or never develops callosal connections, it develops alternative routes to maintain bilateral RSNs and interhemispheric functional connections
  - Data from callosotomy patients suggest that minor commissures and thalamocortical pathways may play a key role in this adaptation.
- demopoulo 2015 : \* conceptual model of ‘sensory processing inefficiency’
  - hyp : impaired interhemispheric communication, mediated by the CC, contributes to information processing inefficiency
    - \* diff to integrate relevant social info
  - the CC may contribute to language learning and cortical development by facilitating rapid and efficient movement of

auditory and visual information to the dominant language-processing hemisphere for integration with past knowledge and interpretation.

## Neuro correlates AgCC

- anatomy tovar-moll2006 :
  - some properties are maintained, other dramatically changed -> compensatory or maladaptative ?
  - pAgCC :
    - \* hyp : genu (anterior) would form first, then more and more posterior -> compatible : remnant CC more genu part
    - \* similar topographic organization as normal CC / remnant genu only connected to frontal (no expansion)
      - in general frontal/genu, parietal/body, occipital/splenium
  - PB : ipsilateral bundle, never seen in controls or in the contralateral hemisphere -> not artefact
    - \* seen in the majority of the cases + preserved topographic organization -> compensatory ?
  - sigmoid bundle : S-shaped, right frontal lobe connected to left occipito-parietal
    - \* 30% of patients, those who are the most severely impaired -> maladaptative ??
  - lazarev 2016 : PB and SB play role in synchrony between some heterotopic regions (EEG coherence)
- functional connectivity (synchronization of signals)
  - hinkley2012 : underconnectivity of some cortical regions (resting state alpha)
    - \* frontal, parietal, occipital -> implicated in altered cognitive functions of AgCC
    - \* still activity in these regions, only less connectivity
    - \* correlated with behavioral measures of executive functions
- structural connectivity (physical pathways)
  - BUT shi2021 : less inter-hemispheric connectivity, more intra-hemispheric connectivity
    - \* Structural alterations not associated with a wide range of neurobehavioral outcomes in children with AgCC
- owen2013 : qualitative organization of resting-state networks is very similar between controls and AgCC
  - reduced inter-hemispheric connectivity, in associative but not in sensory networks
    - \* thalamus could maintain symmetric bilateral connectivity (sensory connections in healthy subjects)
  - Hett2006 : 10% larger anterior commissure on AgCC -> correlated with better cognition in AgCC (Barr2002)
  - high variability in connectivity in AgCC subjects -> find different ways to compensate -> diff to study
- yuan 2020
  - link between structural connectivity (SC) and functional connectivity (FC)
    - \* all brain : faster conduction fibers are likely to produce tighter synchronization of neurophysiological signals
  - Increased intrahemispheric structural/functional connections -> hyp compensatory mechanisms in AgCC ?
  - hyp compensation : could be that AgCC become more lateralized to compensate for bad IH connectivity
    - \* more left-handed in (small) population AgCC and right lateralization of left limbic area
- Szczupak2020 : AgCC affects the interhemispheric synchrony of cortical communication
  - follows developmental gradient : early developed zones (CC develops antero-posterior and medio-lateral) less affected
  - posterior level of sensory areas (connect homotopically) affected
  - cingulate cortex (pioneer axons) synchronized but with a lag
- myelinated axons from the CC link homotopic regions, terminate in cortex where can activate/not GABAergic neurone
  - handle balance E/I -> can inhibit other hemisphere -> controls lateralization and synchronization of brain regions

## Discussion

- reduced axonal density ?
- reduction in axonal count = global property
- no differences in FA/MD -> FBA more sensitive ?

## Limits in studies

- results differ between normalizing CC size by brain size
- when recruiting AgCC participants, select the lower functioning because they have been diagnosed -> higher occurrence of ASD ?

## Different ages in studies

### subregions

- 7 subregions Witelson method
- in some studies, only anterior, posterior, middle

## Evaluation of cognitive functions -> create subgroups ?

### Genetics

- tick 2015: meta-analysis of twin studies on ASD
  - correlations for monozygotic twins (MZ) : .98
  - dizygotic twins (DZ) : 0.53
  - The meta-analytic heritability estimates were substantial: 64–91%
  - meta-analysis shows that extreme cases do not have bigger heritability -> rather continuum than distinct traits
- lau2012 : in individuals with AgCC, we observe that their structural abnormalities are often associated with de novo chromosomal mutations and that the recurrence risk for families is quite low, suggesting that the behavioral phenotypes may also be discontinuous (Sherr et al. 2005)
- lau2012 : common ASD AgCC genetic disorder -> mutations ARX gene (disrupted connectivity and neuronal excitation)
  - shell2003 : ARX (Aristaless Related Homeobox) : critical for brain and neuron development
    - \* have been shown to be responsible for diffuse central nervous system malformations
    - \* autism, AgCC, but also dystonia, epilepsy, thalamic dysplasia, West syndrom...
  - Shell2005 : often de novo mutations
    - \* insertions/deletions identified by CGH array (comparative genomic hybridization)
    - \* ARX males have often more severe symptoms (no other chromosome X to compensate)
    - \* phenotype depends on the genetic dosage (number of copies of a gene, 1 if deletion, 3 if insertion...)
- Belmonte2004 : protein CMH responsible for neural elagage altered in ASD -> cerebellum altered (Purkinje cells) -> social deficits
- Rodriguez2021:
  - three groups of genes that may be involved in the pathophysiology of ASD:
    - \* (1) neurotransmitter receptors (ex. : sérotonine, dopamine, GABA...)
    - \* (2) genes involved in neurotransmitter metabolism (ex. : tryptophane → sérotonine, mélatonine)
    - \* (3) cell adhesion molecules and scaffolding proteins (adhésion cellulaire, échafaudage)
- Wahlsten 2003 : mouse model to explain AgCC :
  - BTBR T/+ tf/tf has AgCC + as well as severe reduction of the hippocampal commissure
    - \* croisement entre gène T hétérozygote et mutation tufted homozygote

### Gender differences

- more prevalent on males
- diagnostic or real gender differences ?

### Role of the CC

- banich 1998
  - CC often conceived as a mechanism for transferring sensory information and coordinating processing between the hemisphere
  - also role in attentional processing, because allows division of labor between hemispheres (more computational power)
    - \* dynamic interactions increase the processing capacity of the brain
    - \* outcome depends on the balance between a better computational power and computational cost for integration info
    - \* 2 hyp : either “open a gates” when processing capacity exceeding in one hemisphere, or CC attentional module that affects attention
  - in 2 attention conceptions :
    - \* as a resource
    - \* as a selective mechanism for gating sensory information
- yuan 2020 :
  - Increased intrahemispheric structural/functional connections -> compensatory mechanisms in AgCC ?
  - hyp compensation : could be that AgCC become more lateralized to compensate for bad IH connectivity
- hinkley 2016 :
  - pAgCC : bilateralization of language / cAgCC right lateralization of language
  - loss left lateralization directed linked to low VIQ -> clinical consequences for atypical lateralization
  - some regions not directly linked to language less activated in the LH
    - \* compensation strategy would come at the expense of processing in these regions
  - supports hyp role CC : transmit information across cortical fields to suppress one area while the contralateral hemisphere is active
- karolis 2019 :
  - hyp lateralization (inter-hemispheric independence hypothesis) : during evolution, brain size expansion led to func-

tional lateralisation in order to avoid excessive conduction delays between the hemispheres -> functionally lateralised regions will be connected less strongly via the corpus callosum than non-lateralised regions to make processing of lateralised functions more efficient.

- but limits recuperation capacity after brain lesions
- gazzaniga2000 : role of the CC in lateralization -> evolutionary theory
  - new function (language) in one hemisphere, the other continues to perform old functions
    - \* lateralization = dev of new skills, maintenance of others
    - \* non-lateralized = more efficient, parallel processing thanks to redundancy (but needs coordination/integration from CC)
  - the CC has enabled the specialization of systems (some of them are lateralized, some of them still have redundancy) :
    - \* LH : language and problem-solving, RH : facial recognition, attention monitoring
    - \* memory : LH more memory of words, RH more memory of faces, objects for both -> more stimulus than memory type (sem/epis)
    - \* attention : endogenous lateralized LH / exogenous independantly in each hemisphere
    - \* coordination : split-brain can perform independant tasks with 2 arms, 2 visual searches, but not 2 decisions (taken successively)
      - advantage for split brain for some simple tasks (parallel performing) but more complex tasks need integration of info
      - can have 2 independant motor plannings (no need for coordination)
      - some types of attention (visual search) non-lateralized and faster when stim divided in the 2H , decision-taking 1 att center
      - but complex tasks (coordination, complex pbs) performed using one attention center even for split-brain, in subcortical areas ?
- monteiro 2019 : lateralization of activation (language and S1 and S2) occurs in AgCC, no symetric activation -> aberrant bundles ?
  - lateralization is not an explication for the absence of split-brain syndrom
- szczupak2022 : new technics that enables to better detect sparle/less dense heterotopic connections -> underestimated!
  - 75% of heterotopic connections in the mice brain, principally in the central core of the CC <-> homotopic periphery
    - \* more numerous in brains with malformations (probst bundles) -> were considered atypical
    - \* heterotopic connections are compensatory connections ?

## Description of CC agenesis

- Glass 2008 :
  - prevalence 2/10000,
  - frequent associated Central Nervous system malformations (49%) and social impairments of autism etc.,
  - 17% chromosomic malformations
  - maternal age at risk, frequently associated with premature
- Paul2014 : overlap ASD AgCC, particularly in communication
- Bedeschi2006 :
  - numbers of many studies ASD among AgCC and contrary
  - no unique prognosis, from normal cognitive level to severe psychomotor delay
  - partial/complete AgCC have similar amount of other cerebral malformations, but different type
    - \* partial AgCC -> correlated with posterior fossa malformations
    - \* complete AgCC -> more frequently associated with malformations of cortical development
  - frequent association with craniofacial abnormalities, genetic anormalities, epilepsy..
- lau2012 : AgCC deficit in social skills, imagination, and mind-reading, communication, attention switching, but not attention to detail
  - intermediate scores between controls and ASD in all domains
  - badaruddin2007 : ACC less severe symptoms than autism
- paul 2014 : problems in attention (mostly), social function, thought, and somatic complaints, no specific interest/attention to detail

The cores syndrome includes: (brown2019)

- (1) Reduced interhemispheric transfer of sensory-motor information;
  - interhemispheric integration of easy stimuli, but not more difficult tasks
  - deficit in bimanual coordination (mueller, 2009, goojiers2013)
    - CC develops slowly + bimanual coord siminal for children/AgCC -> suggests CC essential for bimanual coordination
    - performance on (complex) asymmetric angles correlated significantly with age, independantly of simple motor coordination X age
      - \* IH integration even more essential for complex tasks

- tactual-spatial information (dunn 2000)
  - worse visual and auditory memory processing (paul2016)
    - diff to organize newly learned information for better encoding (LH/RH integration ?)
    - can also be explained by resource limitation (point 3)
  - difficulties encoding verbal and visual information in memory and spontaneously retrieving newly learned information (erickson2014)
    - deficit initial learning and delayed memory
    - the Hemispheric Encoding and Retrieval Asymmetry (HERA) model suggests that the right prefrontal cortex is activated to a greater extent during memory retrieval, whereas the left prefrontal cortex is more activated during encoding of episodic memory
    - commissurotomy -> possible encoding And retrieving in each hemisphere without CC
      - \* weaker memory = consequence of poor integration ?
    - one interpretation of the results reported herein is that verbal learning and memory are deficient due to an inability to enhance the serial word memory expressed by the left hemisphere with the broader semantic fields of the right hemisphere
- (2) Reduced cognitive processing speed;
- differences in cognitive control primary resulting from low processing speed (marco2012)
  - exacerbated when higher task demand (mathias 2004)
- (3) Deficits in complex reasoning and novel problem-solving.
- interhemispheric resources are recruited to assist with cognitively complex tasks (koivisto2000)
  - schieffer2000 : impaired for more complex tasks
    - impairment in abstract reasoning (brown2000), concept formation (fischer1992), problem-solving/formulating strategies (brown2012), generalization (solursh1965)
  - benefit of interhemispheric processing decreases with practice (cherbuin2005)
    - no deficits in overlearned cognitive processes (erickson2013)
  - deficit in creativity : coherent stories but low diversity (brown 2024)
    - model of creativity : interhemispheric inhibition (diversity of cognitive perspectives) then IH integration (creative ideas)
  - poor capacity to integrate contextual information and elaborate imaginative hypotheses (sense of a word) (Su, 2023)
    - diff in generating stories/meaningful cognitive associations to connect pair of words (turk2010)
      - \* -> due to poor integration verbal/semantic LH/RH ?
      - \* link to memory (paul2016)
- (4) hyp : difficulty to understand complex information/integrate sensory/emotional/cognitive information -> difficulty in social behavior
- diff in understanding emotions/language with emotion
    - difficulties in identifying and describing feelings (paul, 2021)
      - \* but similar levels of emotional experience and emotional coping (paul2021)
      - \* impaired emotional interpretation with intact emotional experience -> impaired cognitive processing of emotions
    - deficit recognition complex emotion (fear, anger) (bridgman 2014)
      - \* expressing emotions in words (pazienza2011)
      - \* lack of attention on the eyes
    - deficit recognition of faces (complex material!) (paul2016)
    - basic understanding of emotions but no integration in nuanced social contexts to act in an appropriate way (anderson2017)
      - \* difficulty to reason abstractly about emotions in social context
    - diff to integrate cognitive and emotional info in social situations + diff to anticipate consequences of actions (young2019)
    - diff in imagining and inferring the mental, emotional, and social functioning of others (kang2009)
    - deficit in judgement of appropriate social norms (brown2021)
    - bad at emotional prosodic analysis and nonliteral language interpretation, literal language processing preserved (paul2003)
      - \* bad proverb/non-literal comprehension (rehmel2016)
      - \* emotional prosody requires information from both hemispheres (both LHDamage and RHD have deficits)
      - \* bad integration of HD info (face recognition/emotional) and HG (analytic processing/details)
      - \* deficits especially seen in more complex tasks (require more integration)
  - diff in acting
    - deficit in adaptative behavior, particularly social skills (haisley, 2025, Miller, 2024)
    - rigid application of social norms /no use of context to adapt (brown2021)

- \* lack of flexibility (marco2012)
- low planification, organisation and cognitive flexibility (mangum2021)
- diff to build a strategy/less coherence in choices/diff use available info to take a good decision/influenced by recent choices (brown2012)
- under uncertainty, too much attention to loss (brown 2012)
- more impulsivity/ less inhibition (tendency to answer precipitated or inappropriate)
- consequences for social behavior

#### (4) Others

- low self-awareness of difficulties/metacognition (mangum 2018/2021, kaplan2012)
  - lau2012 : as ASD, parents thought child has greater deficit than he thought he had
- schieffer2000 : difficulties only with older children
- deficit in sustained attention/lack of inhibition (brown2020)
  - mostly on adolescents
    - \* -> adults compensated ?
    - \* or maybe direct consequence of processing speed deficit only for adolescents because of the diff of the task
  - diff sustained conversation, but no lack of interest (paul2014)

### Link between ASD and AgCC

- AgCC brains can compensate from the very beginning, in ASD arrives later, less plasticity
- genetics :
- connectivity :
  - alteration of inter-hemispheric connectivity
    - \* AgCC : Reduced interhemispheric transfer of sensory-motor information
    - \* ASD :
- social
  -

### Cognitive science in general

#### Discourse theory (mason 2006)

- dynamic recruitment of cortical networks in response to text constraints
  - old discourse theories (coarse coding, construction integration framework, salience hypothesis...)
  - 3 theories role RH
    - the spillover of processing to other differential specialized RH networks in response to LH capacity utilization
    - RH used during integration of inferences with the text
    - coarse coding theory : inferences result from activation of coarsely coded semantic information in the RH. Parallel Networks of Discourse:
1. Basic LH Sentence Network (Not Uniquely a Part of Discourse Processing) -> several LH regions
    - syntactic structure, word meaning, visual/graphemic, phono and lexical-semantic processing
  2. A coarse semantic processing network (right middle and superior temporal)
    - activation of coarse meaning and inferences for each word, looks for overlap between other activated
  3. A coherence monitoring network (bilateral dorsolateral prefrontal)
    - increase activity when lack of coherence or unusually semantic field -> helps finding coherence / inferences to integrate
  4. A text integration network (left inferior frontal-left anterior temporal)
    - helps construct the “situation/mental model of the text”
  5. A network for interpreting a protagonist’s or agent’s perspective (bilateral medial frontal/posterior right temporal/parietal)
    - understanding other’s plans/motivations / emotion-based memories
  6. A spatial imagery network (left dominant, bilateral intraparietal sulcus)
    - high-imagery sentences or metaphors

## ideas of relevant analyses

- deficits specific to one of the categories / only score in the 2 categories
  - motor deficit could differ greatly
  - every psycho test can be compared between the 2 categories
  - genetics
  - psycho etc etc
- hypotheses regarding the 2 hypotheses
  - AgCC : primary deficit = disconnection between hemispheres
    - \* motor, coordination
  - ASD : primary deficit = local hyperconnectivity/global hypoconnectivity
    - \* preference local/global, hypersensitivity..
    - \* hypersensitivity linked to attention -> in AgCC ??
- correlations between several “categories” of deficits
  - hilton2007: correlation between social skills and sensory deficits -> correlation only in ASD but not in AgCC ?
  - social skills/motor skills
  - gomes2008: hypersensitivity correlated with attention
  - executive functions
  - dwyer2014 : hyperfocus correlated with quality of life
- neuro : correlation morphometry/fiber density -activations/test results
  - alexander2007 : processing speed (bimanual coordination, other tests...) /morphometry / DTI (FA, MD..)
  - hinkley2012 : rs functional connectivity/executive functions
  - mason2008 : FC between frontal/parietal correlated with size of the CC genu
  - depreto2005 : activity NMS correlated with social ability
  - diamond2019 : social impairment/size of the splenium
  - hett2006 : size of the anterior commissure / “cognition” in AgCC
  - lefevre2014 : intelligence/brain volume
- review language Julio
  - connexion (structural connectivity DWI) of different part of Broca to different part of frontal cortex
    - \* anterior : semantics
    - \* posterior : phonology, syntax, motor coordination
  - AgCC dyspraxia : disruption of salience and fronto-parietal network, symptom disappears, diff disappears
    - \* split-brain “against the patient will”
  - semantics -> temporal, phonology -> frontal
  - vandermerwe2021 :
    - \*
- A four-level model for the characterization of pathological speech
  - sensorimotor control (linguistic-symbolic planning) : temporal-parietal, Broca,
    - \* sound map : left ventral premotor cortex
  - speech-motor planning (sequential organization of movements for each sound + co-articulation (frontal cortical areas))
    - :
    - \* inverse models generate motor commands / forward model predicts consequences of the motor commands -> inability to form sounds/ start-restart behavior ?
    - \* cortical areas in the dominant hemisphere : prefrontal cortex, Area 6, supplementary motor areas, 5 and 7 (posterior parietal), Broca and Wernicke
  - speech motor programming (specification of muscle’s movements, velocity etc + suprasegmental/emotional information) -> duration atypical ?
    - \* rapid compensatory speech movements / auditory feedback
    - \* cortical-subcortical circuits : sensorimotor cortex, lateral cerebellum, basal ganglia
  - speech execution : descending instructions to the lowest levels of control
- AOS : lesion cortical motor areas responsible for planning
  - left ventral premotor cortex ?
  - disruption of both motor planning and/or programming (diff to differentiate) : deficient prosody, slow speech rate, accuracy of articulation, syllabic speech
- summary of zones of interest
  - structural connectivity of different part of Broca to different part of frontal cortex -> diffusion
  - dyspraxia : frontoparietal network -> rs-fmri
  - semantics -> temporal, phonology -> frontal
  - sound map : left ventral premotor cortex -> rs (correlation with SMA, Broca..)
  - speech motor planning : cortical areas in the dominant hemisphere : prefrontal cortex, Area 6, supplementary motor areas, 5 and 7 (posterior parietal), Broca and Wernicke

- speech motor programming : cortical-subcortical circuits : sensorimotor cortex, lateral cerebellum, basal ganglia

## Summary key aspects of ASD

- A variety of symptoms :
  - Many shared with AgCC
    - \* Communication and social interaction
    - \* Attention switching disorder / overselective attention
    - \* Difficulty in novel and complex tasks
    - \* Lack of imagination
  - More specific to ASD
    - \* Restricted and repetitive behaviors
    - \* Hypersensibility
    - \* productive prosody in ASD only ?
- No known physical/anatomical cause

### cognitive theories :

- deficit in executive functions
  - could come from underconnectivity fronto-parietal
- weak central coherence : cognitive style focused on details
  - no global vision or context
  - no flexible integration of various sources
- complexity theory : deficit in integrative context, but not in associative context
  - natural consequence : focus on details
- monotropism : low range of interest
  - competition for attention
  - hyperawareness within the attentinal tunnel, hypoawareness outside it
- hyper-systematization theory : difficulty to cope with high variance systems -> needs good integration

### summary in one neuro-oriented theory

- local overconnectivity / global underconnectivity

## Summary key aspects of AgCC

The cores syndrome includes (not independant) : (brown2019)

### (A) more specific to AgCC, less frequent in ASD

- (1) Reduced interhemispheric transfer of sensory-motor information
  - motor : good integration for easy stimuli, not for more difficult tasks
    - deficit in bimanual coordination, tests with various bi-modalities
  - memory : difficulties encoding verbal and visual information in memory and spontaneously retrieving newly learned information (erickson2014)
    - possible interpretation : verbal learning and memory deficient due to inability to use RH semantic information to enhance LH word memory
- (2) Reduced cognitive processing speed
  - differences in cognitive control primary resulting from low processing speed (marco2012)
  - exacerbated when higher task demand (mathias 2004)

### (B) less specific to AgCC, mostly shared with ASD

- (3) Deficits in complex reasoning and novel problem-solving
  - cognitively complex task : impaired in AgCC (Schieffer2000), interhemispheric resources recruited (koivisto2000)
    - impairment in abstract reasoning (brown2000), concept formation (fischer1992), problem-solving/formulating strategies (brown2012), generalization (solursh1965)
    - novel task : no deficits in overlearned cognitive processes (erickson2013), benefit of interhemispheric processing decreases with practice (cherbuin2005)
  - creativity : diff generating stories (turk2010), coherent stories but low diversity (brown 2024)
    - due to poor integration verbal/semantic LH/RH ?

- model of creativity : interhemispheric inhibition (diversity of cognitive perspectives) then IH integration (creative ideas)
- lack of attention
- (4) hyp : difficulty to understand complex information/integrate sensory/emotional/cognitive information -> difficulty in social behavior
- diff in understanding emotions/language with emotion
  - intact emotional experience and emotional coping (paul2021)
  - impaired emotional interpretation/cognitive processing ->
    - \* identifying
    - \* describing
    - \* integration cognitive and emotional info/abstract reasoning in nuanced social contexts/ judge appropriate social norms
    - \* deficit recognition of faces (complex material!) (paul2016)
    - \* diff in imagining and inferring the mental, emotional, and social functioning of others (kang2009)
  - intact literal language processing
  - impaired emotional prosodic analysis, nonliteral/proverb language interpretation
    - \* emotional prosody requires information from both hemispheres (both LHDamage and RHD have deficits)
    - \* bad integration of HD info (face recognition/emotional) and HG (analytic processing/details)
    - \* deficits especially seen in more complex tasks (require more integration)
- diff in acting
  - deficit in adaptative/flexible behavior : rigid application of social norms /no use of context to adapt (brown2021)
    - \* bad use of available info to take a good decision
    - \* more impulsivity/ less inhibition (tendency to answer precipitated or inappropriate)
    - \* low planification, organisation
  - no lack of interest, but diff sustained conversation

## Why are they common symptoms between ASD and AgCC

- AgCC : no CC -> connectivity deficit, mostly inter-hemispheric
- ASD : local hyperconnectivity -> global underconnectivity
- both have connectivity deficits / difficulty in integrating information from different regions

## Hypothesis about common symptoms

- mostly in the social domain / complex tasks (because social tasks are complex..)
- mostly due to bad integration of information from different sources
- integration deficit between LH and RH
  - LH: lexical, syntax
  - RH : use of context for semantics, emotion recognition
  - discourse theory (mason2006)
    - \* LH sentence network
      - syntactic structure, word meaning, visual/graphemic, phono and lexical-semantic processing
    - \* RH Coarse semantic processing network
    - \* LH test integration network
    - \* bilateral network for interpreting agent's perspective
  - hyp : structural/functional integrity of fronto-parietal network (low size, low connectivity rs)
  - hyp : deficits in
    - \* contextual semantics
    - \* emotions
    - \* multimodal integration
- hyp to be continued ...

## ASD-specific (non-AgCC) inspired hypothesis

- could be linked to local hyperconnectivity
- connectivity deficits in 4 networks (elton2015):

- the default-mode network -> preserved in AgCC
- the dorsal attention network,
- the salience network,
- the executive control network (or fronto-parietal network) -> th ASD = deficit executive function
- monotropism account of ASD (dwyer2024) :
  - \* atypical regulation of attention rather than deficit, both increased endogenous hyper-focus and increased susceptibility to exogenous attention capture
  - \* DAN and VAN overactivated during task-related fmri, giving less flexibility to ECN
- hyp : relatively preserved in AgCC, contrary to ASD
- hypersensitivity and attention
  - in ASD : attention linked to hypersensitivity -> in AgCC ?
    - \* hyp: overselective attention/repetitive behavior compensation mechanisms (liss2006) -> predictable/repetitive events moderate arousal level
  - connectivity of the dorsal attention network/ventral attention network
  - size of the parietal cortex
  - hyp : no hypersensitivity in AgCC / lower correlation with attention ?
- cerebellum smaller in ASD, implicated in inhibition/excitation ratio -> less inhibition
  - Abnormal Cortico-Cerebellar Functional Connectivity with regions related to functions affected in ASD (social cognition)
  - should not be the case in AgCC ? to verify
  - in AgCC, higher cerebellum activation in rs-fmri (siffredi2021) -> alternative pathway
- size/DTI of the CC (for ASD and partial)
  - in general frontal/genu, parietal/body, occipital/splenium
    - \* hyp : genu (anterior) would form first, then more and more posterior -> compatible : remnant CC more genu part
    - \* TODO different functions
  - FC between frontal/parietal correlated with size of the CC genu
  - negative correlation social impairment / fiber density in the CC splenium
  - AgCC : underconnectivity fronto-parietal -> executive functions
    - \* normal activation of each region

## AgCC-specific inspired hypothesis

- hyp comes from disrupted inter-hemispheric communication
- motor deficit
  - motor coordination, integration of stimuli in different modalities
  - hyp : under-connectivity motor regions left/right (M1, SMA, parietal..) + correlated with bimanual tasks
  - hyp : higher connectivity in ASD, and for more complex tasks, higher correlations connectivity/score
- low processing speed
  - lower homotopic connectivity ?? (exploratory)
  - ASD and partial : PS correlated with integrity of callosal fibers
  - Processing speed correlated with integrity of WM tracts in parietal / temporal cortices / left middle frontal gyrus (turken2008)
- lateralization (hinkley2016)
  - rs-fmri : intra-hemispheric connectivity VS inter-hemispheric connectivity